

Vineyard monodromy in Newton's equations

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Looking for vineyard monodromy in the wild, I found it in classical mechanics, in the figure-eight solution of the three-body problem. What it measures is the more interesting half.

Three equal masses chase one another around a figure-eight curve, the orbit Moore found numerically and Chenciner and Montgomery proved exists. Sum a radial kernel at each body and follow the degree-zero persistence of its superlevel sets over a third of the period, the essential class dying at the infimum. The diagram returns to itself with its three points cyclically permuted, so the monodromy has order 3. Those three features stay clear of the diagonal throughout, the least persistent at 0.06, and are distinct except at the wall crossings, where two coincide exactly; following the critical points through shows them exchanged, not turned back, so the cycle is transport, not a relabeling of coincident points. Ordering strands by birth at the first slice, running time forwards and signing each crossing by the death coordinate, the braid is $(\sigma_2^{-1}\sigma_1)^2$, whose trace closure is the figure-eight knot, by its Alexander polynomial $t - 3 + t^{-1}$ and independently by SnapPy.

The loop is $T/3$ because after a third of a period the bodies occupy the same three positions with their labels shifted, and the density is a *sum* over them, so relabeling leaves it unchanged as a function and the diagram returns exactly, not nearly. Scanning for set-returns finds only $T/3$ and $2T/3$, so $T/3$ is primitive; over the full period the relabeling is the identity and the monodromy trivial. Nothing about the loop is chosen.

What orders the vines is easy to write down. At the bodies $h_i = \sum_j K(r_{ij})$, the shared term cancels, and $h_1 - h_2 = K(r_{13}) - K(r_{23})$ vanishes when body 3 is equidistant from bodies 1 and 2, an *isosceles configuration*. The peaks do the same, and exactly. Let σ be the reflection in the perpendicular bisector of bodies 1 and 2. It fixes that pair, so $\rho \circ \sigma - \rho = K(|x - \sigma q_3|) - K(|x - q_3|)$, which is positive on body 2's side whenever body 3 lies nearer body 1. Then $h_1 \geq \rho(\sigma M_2) > \rho(M_2) = h_2$, the first step because σM_2 sits as far from body 1 as M_2 does from body 2, and ρ is concave on that ball, so M_1 is its highest point; and at equidistance σ carries ρ to itself, so the two are equal. So wherever that concavity holds, the walls are the isosceles configurations and nothing else, and the strands are ordered by the opposite sides. The deaths are the two shortest, the minimum spanning tree, measured rather than proved. The braid reads only the order of the births and of the deaths, never their size.

The condition contains no kernel, with a consequence I did not expect: a Gaussian, a cusped exponential, a Cauchy and a Student t all put the walls at the same twelve instants. Nor can the bandwidth. What it sets is a window: below $\sigma \approx 0.24$ the crowding differences fall towards numerical resolution, above $\sigma \approx 0.30$ a feature meets the diagonal, neither bound topological.

A change of frame, even a time-dependent one, carries $\rho(x, t)$ to $\rho(g_t^{-1}x, t)$. Every superlevel set goes to a congruent one, so each diagram is untouched and the whole vineyard with it. Viewed from six frames rotating at different rates the vineyard braid is $(\sigma_2^{-1}\sigma_1)^2$ every time, while the trajectory braid picks up a twist per turn. The two are invariants of different objects: the trajectory braid of the motion in the plane, the vineyard braid of that motion reduced by translations and rotations.

So the monodromy is no coincidence of this orbit: its order is 3 because the time shift permutes the bodies cyclically, and over $T/3$ the curve crosses four of the twelve walls, the braid being its itinerary through the chambers. And the knot carries content of its own. Over that same $T/3$ the bodies trace the pseudo-Anosov braid $\sigma_1\sigma_2^{-1}$, up to the conjugacy a projection introduces; under the standard map to $SL(2, \mathbb{Z})$ it is $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, whose mapping torus on the once-punctured torus is the figure-eight knot complement, which is why this orbit is known by its braid type. Conjugating the square of that braid by σ_1 returns the vineyard word letter for letter, so the two close to the same knot and the vineyard carries exactly twice the topological entropy, 1.924847 against 0.962424. And unlike the trajectory braid, whose word depends on the projection chosen, the vineyard braid needs none.